



The smartphone uses Google Nexus 5 for the hardware and works out of the box

It is an Android based smartphone that is powered by head-tracking technology. It also packs an advanced computer vision algorithm and a front-facing shooter that helps track the user's head movements and allows the user to control a cursor on the screen. The cursor is like a virtual finger that allows the user to do everything others can do with a regular smartphone.

The developer of the smartphone, Giora Livne, is disabled and he plans to hand out 30 Sesame smartphones to differently-abled people who have been nominated by their peers. Livne conceptualised the smartphone after watching a TV demonstration for a game that was controlled by head movements.

Livne holds a degree in electrical engineering and he estimated correctly the technology's potential to help him and the others. Now he is able to text and send WhatsApp messages to his friends and children on a regular basis. The phone has helped him get rid of social isolation, which affects several disabled people.

Smartphone that can diagnose HIV

A team of researchers from Columbia University, the USA, has developed a low-cost smartphone accessory that can perform a test that simultaneously detects three infectious disease markers from a finger prick of blood in just 15 minutes.

Led by Samuel K. Sia, associate professor of biomedical engineering at Columbia's School of Engineering and Applied Science, the device replicates the mechanical, optical and electronic functions of a lab based blood test. Specifically, it performs an enzyme-linked immunosorbent assay (ELISA) without requiring any stored energy, as all necessary power is drawn from the smartphone.

Early diagnosis and treatment in pregnant mothers can greatly reduce adverse consequences to both mothers and their babies.

The device is small and light enough to fit into a hand and can run assays on disposable plastic cassettes

Now BINAY Human-Centric Circadian PowerLED Lights Can Make You Healthy



BINAY kelvin-changing Human-Centric Circadian PowerLED Lights are designed to affect the human circadian rhythm positively for enhanced physical and mental well being

BINAY has developed kelvin-changing Human-Centric Circadian PowerLED light fittings which have colour temperature changing properties. The colour temperature of the light emitted by these fittings can be manually changed as desired, or set to change according to a pre-set programme. These lights affect the body clock in the human body positively, encouraging the release of appropriate hormones depending on the time of the day (thus leading to increased alertness and productivity during daytime, and improved mental and physical relaxation in the evening).

Greatest importance for applications in:

- **Workplaces** such as office and factories (for enhanced productivity and improved alertness)
- **Healthcare facilities** such as hospitals and nursing homes (to facilitate patient recovery)
- **Educational applications** such as school and colleges (where enhanced mental concentration is required)
- **Residential applications** (alert and productive cool daylight during the day, and restful warm white in the evening)



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